

Project Initialization and Planning Phase

Date	4 July 2026
Team ID	XXXXX
Project Title	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This project aims to predict the **Human Development Index (HDI)** of a country using **Machine Learning**. The system analyzes important development indicators such as **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) Per Capita** to generate an accurate HDI score and category. The solution improves prediction speed, reduces manual effort, and supports informed decision-making.

Project Overview	
Objective	To develop a machine learning-based web application that predicts the Human Development Index (HDI) of a country using key socio-economic indicators, providing quick and accurate predictions.
Scope	The project collects HDI-related data, preprocesses it, trains a Linear Regression model, and integrates it with a Flask web application to provide instant HDI predictions through an easy-to-use interface.
Problem Statement	
Description	Predicting the Human Development Index manually is difficult because it requires analyzing multiple indicators such as life expectancy, education, and income. Manual calculations are time-consuming and may lead to errors.
Impact	An automated HDI prediction system improves accuracy, saves time, and assists governments, researchers, and policymakers in analyzing development levels and making informed decisions.
Proposed Solution	
Approach	The project uses the Linear Regression machine learning algorithm to train a prediction model using historical HDI data. The trained model is saved using Pickle and integrated into a Flask web application, allowing users to enter values and instantly receive the predicted HDI score and category.
Key Features	- Machine Learning-based HDI prediction. - Predicts HDI using four input features. - User-friendly Flask web application.

	- Fast and accurate prediction results. - Displays both HDI score and HDI category. - Simple interface built using HTML and CSS.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 or equivalent
Memory	RAM	8 GB
Storage	Disk space	256 GB SSD or above
Software		
Frameworks	Web Framework	Flask
Libraries	Machine Learning & Data Processing	scikit-learn, pandas, numpy, matplotlib
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, format	HDI Dataset (CSV format)